

## SAFETY DATA SHEET

Revision Date 29-March-2024

Revision Number 4

### 1. Identification

**Product Name** o-Toluidine

**Cat No. :** A13602

**CAS-No** 95-53-4  
**Synonyms** 2-Aminotoluene; 2-Methylaniline

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                   |             |
|-----------------------------------|-------------|
| Flammable liquids                 | Category 4  |
| Acute oral toxicity               | Category 3  |
| Acute dermal toxicity             | Category 4  |
| Acute Inhalation Toxicity         | Category 3  |
| Skin Corrosion/Irritation         | Category 2  |
| Serious Eye Damage/Eye Irritation | Category 1  |
| Germ Cell Mutagenicity            | Category 2  |
| Carcinogenicity                   | Category 1A |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Combustible liquid  
Harmful in contact with skin  
Toxic if swallowed or if inhaled  
Causes skin irritation  
Causes serious eye damage  
Suspected of causing genetic defects  
May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic organisms  
Toxic to aquatic life with long lasting effects  
Light sensitive

### 3. Composition/Information on Ingredients

| Component              | CAS-No  | Weight % |
|------------------------|---------|----------|
| Benzenamine, 2-methyl- | 95-53-4 | <= 100   |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical

advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms/effects**

Difficulty in breathing. Causes eye burns. Causes severe eye damage. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media**Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.**Unsuitable Extinguishing Media**

No information available

**Flash Point**

85 °C / 185 °F

**Method -**

No information available

**Autoignition Temperature**

480 °C / 896 °F

**Explosion Limits****Upper**

7.50%

**Lower**

1.50%

**Sensitivity to Mechanical Impact** No information available**Sensitivity to Static Discharge** No information available**Specific Hazards Arising from the Chemical**

Combustible material. Flammable. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA****Health**  
3**Flammability**  
2**Instability**  
0**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.

## 7. Handling and storage

### Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

### Storage.

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Protect from direct sunlight. Keep locked up. Keep under nitrogen. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component              | Alberta                                          | British Columbia   | Ontario TWA/EV     | Quebec                                           | ACGIH TLV          | OSHA PEL                                                                                                       | NIOSH        |
|------------------------|--------------------------------------------------|--------------------|--------------------|--------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------|--------------|
| Benzenamine, 2-methyl- | TWA: 2 ppm<br>TWA: 8.8 mg/m <sup>3</sup><br>Skin | TWA: 2 ppm<br>Skin | TWA: 2 ppm<br>Skin | TWA: 2 ppm<br>TWA: 8.8 mg/m <sup>3</sup><br>Skin | TWA: 2 ppm<br>Skin | (Vacated) TWA: 5 ppm<br>(Vacated) TWA: 22 mg/m <sup>3</sup><br>Skin<br>TWA: 5 ppm<br>TWA: 22 mg/m <sup>3</sup> | IDLH: 50 ppm |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Ammonia and organic ammonia derivatives filter Type K Green conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                               |
|----------------------------------------|-------------------------------|
| Physical State                         | Liquid                        |
| Appearance                             | Light yellow                  |
| Odor                                   | Rotten-egg like               |
| Odor Threshold                         | No information available      |
| pH                                     | 7.5 8 g/l aq. sol             |
| Melting Point/Range                    | -23 °C / -9.4 °F              |
| Boiling Point/Range                    | 199 - 200 °C / 390.2 - 392 °F |
| Flash Point                            | 85 °C / 185 °F                |
| Evaporation Rate                       | No information available      |
| Flammability (solid,gas)               | Not applicable                |
| Flammability or explosive limits       |                               |
| Upper                                  | 7.50%                         |
| Lower                                  | 1.50%                         |
| Vapor Pressure                         | 0.3 mbar @ 20 °C              |
| Vapor Density                          | 3.7 (Air = 1.0)               |
| Specific Gravity                       | 1.000                         |
| Solubility                             | No information available      |
| Partition coefficient; n-octanol/water | No data available             |
| Autoignition Temperature               | 480 °C / 896 °F               |
| Decomposition Temperature              | > 200°C                       |
| Viscosity                              | No information available      |
| Molecular Formula                      | C7 H9 N                       |
| Molecular Weight                       | 107.15                        |

## 10. Stability and reactivity

|                                  |                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                   |
| Stability                        | Stable under normal conditions. Light sensitive. Air sensitive.                                                              |
| Conditions to Avoid              | Exposure to air. Exposure to light. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids                                                                                        |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                               |
| Hazardous Polymerization         | No information available.                                                                                                    |
| Hazardous Reactions              | None under normal processing.                                                                                                |

## 11. Toxicological information

**Acute Toxicity****Product Information**

**Component Information**

| Component              | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation            |
|------------------------|--------------------------|------------------------------|----------------------------|
| Benzenamine, 2-methyl- | LD50 = 670 mg/kg ( Rat ) | LD50 = 3235 mg/kg ( Rabbit ) | LC50 = 862 ppm ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component              | CAS-No  | IARC    | NTP   | ACGIH | OSHA | Mexico |
|------------------------|---------|---------|-------|-------|------|--------|
| Benzenamine, 2-methyl- | 95-53-4 | Group 1 | Known | A3    | X    | A3     |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

*A4 - Not Classifiable as a Human Carcinogen*

*A5 - Not Suspected as a Human Carcinogen*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

| Component              | Freshwater Algae                                       | Freshwater Fish | Microtox                | Water Flea |
|------------------------|--------------------------------------------------------|-----------------|-------------------------|------------|
| Benzenamine, 2-methyl- | EC50: = 3.7 mg/L, 96h static (Desmodesmus subspicatus) | Not listed      | EC50 = 13.2 mg/L 30 min | Not listed |

|  |                                                    |  |  |  |
|--|----------------------------------------------------|--|--|--|
|  | EC50: = 3.9 mg/L, 72h<br>(Desmodesmus subspicatus) |  |  |  |
|--|----------------------------------------------------|--|--|--|

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component              | log Pow |
|------------------------|---------|
| Benzenamine, 2-methyl- | 1.4     |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                        | RCRA - U Series Wastes | RCRA - P Series Wastes |
|----------------------------------|------------------------|------------------------|
| Benzenamine, 2-methyl- - 95-53-4 | U328                   | -                      |

### 14. Transport information

#### DOT

UN-No UN1708  
Hazard Class 6.1  
Packing Group II

#### TDG

UN-No UN1708  
Hazard Class 6.1  
Packing Group II

#### IATA

UN-No UN1708  
Proper Shipping Name TOLUIDINES, LIQUID  
Hazard Class 6.1  
Packing Group II

#### IMDG/IMO

UN-No UN1708  
Proper Shipping Name TOLUIDINES, LIQUID  
Hazard Class 6.1  
Packing Group II

### 15. Regulatory information

#### International Inventories

| Component              | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Benzenamine, 2-methyl- | 95-53-4 | X   | -    | X    | ACTIVE                                        | 202-429-0 | -      | -   |

| Component              | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------|---------|-------|----------|------|------|------|------|-------|-------|
| Benzenamine, 2-methyl- | 95-53-4 | X     | KE-23446 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances  
 AICS - Australian Inventory of Chemical Substances  
 PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component              | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                  | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | -                                                                   | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 43.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - Carcinogenic (Article 57a)                                                      |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

#### REACH links

<https://echa.europa.eu/authorisation-list>  
<https://echa.europa.eu/substances-restricted-under-reach>  
<https://echa.europa.eu/candidate-list-table>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component              | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Benzenamine, 2-methyl- | 95-53-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component              | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzenamine, 2-methyl- | 95-53-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

#### Prepared By

Product Safety Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

#### Revision Date

29-March-2024

#### Print Date

29-March-2024

#### Revision Summary

New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other



materials or in any process, unless specified in the text

**End of SDS**